

EFFECTS OF DILSUKH (HERBAL DRUG) ON CARDIAC PARAMETERS

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ABSTRACT

The purpose of this study was to establish the physiological and pharmacological effects of DILSUKH drug on cardiac activity. The two dilutions of DILSUKH drug (25 mg/ml and 50 mg/ml) reduced the heart rate and increased the duration of cardiac cycle significantly while it increased the cardiac force non significantly. It is concluded that DILSUKH drug is physiologically active which regulates cardiac activity by increasing the efficiency of heart and thus claim of practitioners is justified.

INTRODUCTION

Use of herbal medicine has gained world-wide recognition on the basis of their minimum side effects in comparison with synthetic drugs used in allopathic treatment. Although the efficacy of these drugs have been advocated by their practitioners (*Tabibs/Hakims*) but their physiological and pharmacological effects have not been studied on scientific basis.

Hamdard Laboratories (Waqf) Pakistan has been manufacturing and practicing such herbal medicines since long time. These medicines have been prepared from a single herb as well as by mixing the ingredients of many herbals. One of such medicine is DILSUKH. This medicine has been prescribed for the reduction of blood cholesterol and the regulation of heart beat and it includes the ingredients of medicinal plants.

The scientific studies are however available in literature regarding cardiac effects of various plants extracts. The aqueous extract of *Eremophila alternifolia* have been found to exhibit cardio-active effect (Pennacchio, *et al.*, 1995). Similarly, the cardiovascular effects have been demonstrated by using Aqueous extract of *Moringa pterygosperma* (Dnyanesh *et al.*, 1995). Tariq Sharafatullah *et al.* (1985) reported the effect of garlic extract on Myocardial Contraction of rabbit's heart and found its inhibitory action on heart. Similarly, the effects on blood pressure have been reported by Aftab *et al.* (1996) by using crude extracts of *Cassia absus L.*

On the basis of above literature and significance of the extracts of various herbs in terms of their physiological and Pharmacological actions and use of Dilsukh by the practitioner as

regulator of heart beat and having cholesterol reduction property, this study was conducted to establish the effects of dilsukh on cardiac activity.

MATERIALS AND METHODS

Drugs:

Dilsukh (herbal drug) was used in the present study that was obtained from Hamdard Laboratories (Waqf) Pakistan. Its two dilutions of 25 mg/ml and 50 mg/ml were prepared for experimental use. In addition, two dilutions of sympathetic and parasympathetic agents, i.e., Adrenaline (PDH Labs. Pvt. Ltd., Lahore, Pakistan) and Acetylcholine chloride (Merck) were used for the purpose of comparison with the above herbal drug. Fresh dilutions were prepared in 0.65% saline (Isotonic for amphibians) at the time of experiment from refrigerated stock solution of these drugs.

Animals:

In the present study both male and female adult frogs *Rana Cyanophlictis* were used.

Experimental Procedure:

After using the pithing procedure the unconscious frog was fixed on frog board to remove the pericardium of its heart. Exposed heart was then fixed with the heart lever through a nylon thread and hook, inserted at the apex of heart. During experiment, normal Ringer solution (NaCl, 115 mM; KCl, 2.5 mM; CaCl₂, 1.8 mM; Na₂HPO₄, 2.15 mM; Na₂HPO₄ 0.85 mM) was continuously poured on heart to protect its drying. The movement of heart lever (Harvard Apparatus Cat. No.50-0660) was recorded through an ink stylus on Harvard Kymograph (Universal Model Cat. No.50-7392) by moving its drum at the speed of 10 mm/sec. After getting the normal record, the heart was continuously poured by various drug dilutions of sympathetic, parasympathetic drugs and Dilsukh, one by one. Care was taken to pour the drug for atleast 3 to 5 minutes before the recording of its effect. After the recording of effect, the heart was washed with normal Ringer solution for 3 to 5 min. before the other drug was tested.

Measurements:

The records obtained were used for the measurements and calculations of heart rate/(minute), cardiac cycle (sec.), cardiac force (gm), by using kymograph speed (10 mm/sec) and heart lever calibration (2g weight).

Statistical Analysis:

The data was analysed by using standard statistical tools, i.e., mean, Standard Error, and percent control. The comparison of control and treated values was done by using student's t- test and the significance level was considered from p level of 0.05.

RESULTS

The effect of cholinergic, and adrenergic drug dilutions and two concentrations of Dilsukh drug on various cardiac parameters has been shown in Table 1.

1. Effect on Heart Rate:

The effect of 10^{-6} dilution of Ach on exposed heart had demonstrated a decrease in heart rate with respect to its control, this decrease was statistically non significant ($P > 0.05$). However, 10^{-6} and 10^{-5} dilutions of adrenaline had resulted in a non significant ($P > 0.05$) increase in heart rate. On the other hand, the two dilutions of Dilsukh drug, i.e., 25 mg/ml and 50 mg/ml had reduced the heart rate significantly being $P < 0.025$ and $P < 0.005$ respectively.

2. Effect on Cardiac Force:

A dilution of 10^{-6} prepared from Ach demonstrated non-significant ($P > 0.05$) increase in cardiac force. While two dilutions of Adrenaline, i.e., 10^{-5} and 10^{-6} demonstrated reduction in cardiac force which was also non-significant ($P > 0.05$), statistically. On the other hand the two concentrations of Dilsukh drug (25 mg/ml and 50 mg/ml) have resulted in a non-significant ($P > 0.05$) increase in cardiac force.

3. Effect on Cardiac Cycle:

A dilution of 10^{-6} of Ach showed increase in the duration of cardiac cycle which was statistically non-significant ($P > 0.05$). A dilution of 10^{-6} adrenaline also showed an increase in duration of cardiac cycle which was statistically non-significant, ($P > 0.05$) and 10^{-5} dilution of Adrenaline decreased it which was also statistically non-significant ($P > 0.05$). While the two concentrations of Dilsukh drug i.e., 25 mg/ml and 50 mg/ml increased the duration of cardiac cycle which was statistically significant, i.e., $P < 0.025$ and $P < 0.005$ respectively.

DISCUSSION

In the present study the drug **Dilsukh** has been tested for its effects on cardiac parameters that demonstrates it as physiologically active. The results presented above clearly show that the two concentrations (25 mg/ml and 50 mg/ml) of the drug has significantly reduced the heart rate while increased the cardiac force (non-significantly), and cardiac cycle (significantly). It is important to note that in these results, the reduction in the heart rate has been associated with an increase in cardiac force which suggests that this drug is increasing the efficiency of heart by reducing the heart rate, and increasing the duration of the cardiac cycle, significantly (Table 1).

It has been reported that as a result of training in athletes, the resting heart rate decreases (Katona et al., 1982), and thus is responsible for the efficiency of heart during athletic performance. The observed effect of *Dilsukh*, i.e., decrease in the heart rate, and increase in

Table I
A statistical comparison of the effect of various cholinergic and adrenergic drug dilutions and two concentrations of Dilsukh drug on Cardiac parameters. The values have been presented as mean $\pm$ S.E.

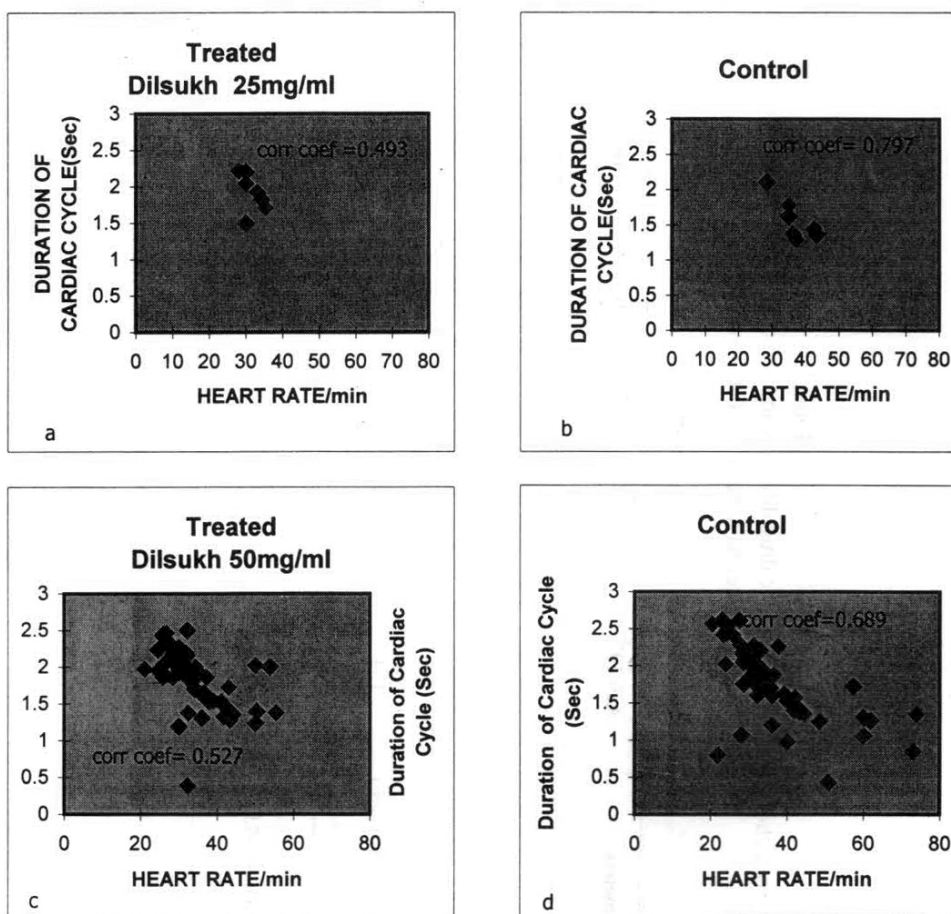
Parameters	Control	Ach 10^{-6}	Control	Adr 10^{-6}	Control	Adr 10^{-5}	Control	Dilsukh 25 mg/ml	Control	Dilsukh 50 mg/ml
Heart rate/mm	44.79 $\pm$ 3.58 (5)	37.72 $\pm$ 1.65 (5)	43.22 $\pm$ 4.63 (5)	40.53 $\pm$ 4.37 (5)	48.46 $\pm$ 2.97 (7)	48.51 $\pm$ 1.89 (7)	36.81 $\pm$ 1.01 (7)	31.52 $\pm$ 1.40 (65)	39.71 $\pm$ 0.89 (65)	34.89 $\pm$ 0.89 (65)
	$p > 0.05$		$p > 0.05$		$p > 0.05$		$p > 0.05$		$p > 0.005$	
Cardiac force/(gm)	0.238 $\pm$ 0.05 (5)	0.248 $\pm$ 0.05 (5)	0.188 $\pm$ 0.035 (3)	0.154 $\pm$ 0.0163 (3)	0.168 $\pm$ 0.034 (4)	0.164 $\pm$ 0.025 (4)	0.387 $\pm$ 0.041 (7)	0.451 $\pm$ 0.053 (7)	0.299 $\pm$ 0.017 (61)	0.302 $\pm$ 0.015 (61)
	$p > 0.05$		$p > 0.05$		$p > 0.05$		$p > 0.05$		$p > 0.05$	
Cardiac cycle/(sec)	1.27 $\pm$ 1.44 (5)	1.54 $\pm$ 0.47 (5)	1.238 $\pm$ 0.066 (3)	1.41 $\pm$ 0.158 (3)	1.56 $\pm$ 0.30 (5)	1.33 $\pm$ 0.07 (5)	1.56 $\pm$ 0.111 (7)	1.91 $\pm$ 0.099 (7)	1.61 $\pm$ 0.060 (66)	1.80 $\pm$ 0.047 (66)
	$p > 0.05$		$p > 0.05$		$p > 0.05$		$p > 0.05$		$p < 0.005$	

Table 2
Effect of various cholinergic and adrenergic drug dilutions and two concentrations of Dilsukh drug on cardiac parameters.
The values have been presented as percent of control.

Parameters	Ach 10^{-6}	Adr 10^{-6}	Adr 10^{-5}	Dilsukh 25 mg/ml	Dilsukh 50 mg/ml
Heart rate/min.	84	93	100	85	88
Cardiac force (gm)	104	82	97	116	101
Cardiac cycle (sec.)	121	101	85	123	111

the duration of the cardiac cycle probably associated with an effect which provide efficient pumping of the heart and thus regulates the (required) output.

FIG.-1 SCATTER PLOT OF RELATIONSHIP BETWEEN HEART RATE AND CARDIAC CYCLE



On the basis of this discussion the claim of practitioners that it regulates the heartbeat appears to be physiologically justified. A comparison of the two concentrations of **Dilsukh** is presented on three cardiac parameters as percent of their control (Table 2). It demonstrates that a dose of 25 mg/ml of **Dilsukh** is more effective than its concentration of 50 mg/ml.

This is an unexpected observation which suggests the need of dose-response study of this drug. Further, the comparison of the effect of this drug with cholinergic and adrenergic agents as shown in Table 2 as percent of their control demonstrated that the physiological effects of Dilsukh more closely resemble with the cholinergic effects that also decrease the heart rate while increasing the duration of cardiac cycle (Table 2).

The effects of cholinergic and adrenergic agents on three cardiac parameters studied demonstrated statistically non-significant results which are probably due to their dilution factor.

An attempt has also been made to compare the relation of the heart rate versus cardiac cycle in control and **Dilsukh** (25 mg/ml and 50 mg/ml) influenced heart (Fig. 1 a, b, c, and d). According to these relationship, **Dilsukh** is physiologically responsible to influence the correlation between the heart rate and cardiac cycle by decreasing correlation coefficient from 0.797 (Control) to 0.492 (with 25 mg/ml Dilsukh) and from 0.669 (control) to 0.527 (with 50 mg/ml **Dilsukh**). Although, the relation between these two parameters is slightly disturbed by both the concentrations of **Dilsukh**, but probably this is the mode of action of **Dilsukh**, thus affecting the cardiac efficiency positively, as evident from the increase in cardiac force though non-significant (Table 1 and 2).

From the above discussion it is concluded that **Dilsukh** is a physiologically active drug which regulates cardiac activity by lowering the heart rate, while increasing the duration of cardiac cycle and cardiac force. However, further studies on mammalian heart are desirable using different concentrations of **Dilsukh** along with the establishment of its receptor activity.

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